RICCARDO CADEI

PhD Researcher - Machine Learning and Causality

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Education

ISTA — Ph.D. Computer Science

Jan 2024 - Dec 2027 [expected] Vienna, Austria

Google PhD Fellowship
Scaling machine learning methods for causal inference to complex measurements in scientific experiments [9, 6, 8, 5, 7, 10] + [13, 14, 15, 16, 17, 18].
Supervisor: Francesco Locatello ***Group***: Causal learning and Artificial Intelligence

INRIA (Paris) — **Visiting**
Modern Computer Vision for scientific experiments understanding.
Supervisor: Cordelia Schmid ***Group***: Willow ***Program***: ELLIS (ELSA)

EPFL — M.Sc. Data Science

Sep 2020 – Feb 2023 Lausanne, Switzerland

Relevant courses: ML, ANN, DL, Applied Data Analysis, Visual Intelligence.
Grade: 5.53/6 ***Teaching Assistant:*** Introduction to Machine Learning

Harvard University (Cambridge, MA) — **Visiting**
Thesis: *Interpretable Het. Treatment Effect discovery and inference* [3, 4] + [a,b].
Grade: 6/6 ***Grant***: Causal Inference for Machine Learning

Politecnico di Milano — B.Sc. Mathematical Engineering

Sep 2017 – Jul 2020 Milan, Italy

Thesis: *Mathematical Programming for activity planning in Oncology Day-Hospital.*
Grade: 110/110 ***Associations***: PoliMi Data Scientists, Ass. Ing. Matematici

Other Research Appointments

Harvard University

Sep 2022 – Sep 2023 Cambridge (MA), United States

Research Assistant @NSAPH: Causal Machine Learning for climate change, environmental impacts on health outcomes, and regulatory policy [3, 4] and [12].

Schlumberger-Doll Research

Feb 2022 – Aug 2022 Cambridge (MA), United States

Machine Learning Researcher: Deep Learning for Causal Modeling and interpretation of acoustic subsurface data for anomaly detection and prevention.

École polytechnique fédérale de Lausanne

Nov 2020 – Feb 2022 Lausanne, Switzerland

Research Assistant (*Summer Intern*) @iGH: Developing a mobile app for (non-invasive) upper body posture detection using Deep Learning.
Research Assistant @*VITA*: Introducing a Causal formalism and a Robust and Adaptive modular architecture for Motion Forecasting [2] and [11].
Research Assistant @LESO-PB: Introducing a U-Net based model for detecting available rooftop areas to install photovoltaic panels from satellite images [1].

Consulting and Entrepreneurship

Entrepreneur First

Oct 2023 – Dec 2023 Paris, France

Founder in Residence: Learning entrepreneurial skills while trying to launch a start-up in Responsible AI and Sustainability at StationF.

L.O.L. Consultants

Dec 2020 – Feb 2021 [remote] Melbourne, Australia

Machine Learning Engineer: Detection of available rooftop area to install photovoltaic panels from high-quality satellite images using Deep Learning.

Awards

Career

Nova 111 Student List 2023
Selected among the 10 most promising Italian Computer Scientists Under25.

Machine Learning

ICLR - Oral 2026
Selected oral presentation (top 1%) for "Exploratory Causal Inference in SAEnce" [9].

Google PhD Fellowship 2026-7
In Machine Learning and ML foundations.

Other scholarships 2025-6
G-Research travel grant, ELSA travel grant.

Jane Warren Award 2023
By Health Effects Institute for "Causal Rule Ensemble" algorithm [3].

Generali Data Challenge 2021
Best model and code in the Churn Classification Datahon at @Generali S.p.a out of 280+ participants.

Higgs Boson Challenge 2020
2nd place in the AICrowd final challenge of Machine Learning course at @EPFL out of 290+ teams.

Oracle GraphML Contest 2019
1st place in the Kaggle final challenge of Graph Machine Learning course at @Politecnico di Milano in partnership with @Oracle Labs.

Mathematics

International competition for mathematical and logical games 2018
5th national place (ITA), class L2 (Under21).

Grand Prix of Applied Mathematics
5th national place (ITA) out of 7500+ students. 2017
6th national place (ITA) out of 7500+ students. 2016

Coding

Machine Learning: Python , R , Julia
Deep Learning: PyTorch , Tensorflow , HuggingFace
Math: MATLAB , Python , R , AMPL
Big Data: Spark , Scala , SQL , HDFS , AWS
Robotics: RobotC , C , Python
App and Web: HTML , CSS , Android Studio

Languages

Italian: C2, **English**: C1, **French**: A1

Referees

Dr. Francesca Dominici — **Harvard**
@ fdominic@hsph.harvard.edu

Dr. Francesco Locatello — **ISTA**
@ francesco.locatello@ist.ac.at

Dr. Cordelia Schmid — **Google/INRIA**
@ cordelias@google.com

Publications

Google Scholar statistics
Total citations: 301
h-index: 6

Full Articles (arxiv, main conferences, journals), in blue two selected publications

[1] Roberto Castello, Alina Walch, Raphael Attias, **RC**, Shasha Jiang, and Jean-Louis Scartezzini. Quantification of the suitable rooftop area for solar panel installation from overhead imagery using Convolutional Neural Networks. In ***Journal of Physics**: Conference Series*, volume 2042, page 012002. IOP Publishing, 2021.

[2] Yuejiang Liu, **RC**, Jonas Schweizer, Sherwin Bahmani, and Alexandre Alahi. Towards Robust and Adaptive Motion Forecasting: A Causal Representation Perspective. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, pages 17081–17092, 2022.

[3] Falco J Bargagli-Stoffi*, **RC***, Kwonsang Lee, and Francesca Dominici. Causal rule ensemble: Interpretable Discovery and Inference of Heterogeneous Treatment Effects. *[**under review at Journal of the Royal Statistical Society: Series A**]*, 2023.

[4] **RC***, Naeem Khoshnevis*, Kwonsang Lee, Daniela Maria Garcia, and Falco J. Bargagli-Stoffi. CRE: an R package for interpretable discovery and estimation of Heterogeneous Treatment Effect. ***Journal of Open Source Software***, 2023.

[5] **RC**, Lukas Lindorfer, Sylvia Cremer, Cordelia Schmid, and Francesco Locatello. Smoke and mirrors in causal downstream tasks. *Advances in Neural Information Processing Systems (**NeurIPS**)*, 2024.

[6] **RC**, Ilker Demirel, Piersilvio De Bartolomeis, Lukas Lindorfer, Sylvia Cremer, Cordelia Schmid, and Francesco Locatello. Causal lifting of neural representations: Zero-shot generalization for causal inferences. *Advances in Neural Information Processing Systems (**NeurIPS**)*, 2025.

[7] Dingling Yao, Shimeng Huang, **RC**, Kun Zhang, and Francesco Locatello. The third pillar of causal analysis? a measurement perspective on causal representations. *Advances in Neural Information Processing Systems (**NeurIPS**)*, 2025.

[8] Dingling Yao, Dario Rancati, **RC**, Marco Fumero, and Francesco Locatello. Unifying Causal Representation Learning with the Invariance Principle. *Proceedings in the Thirteenth International Conference on Learning Representations (**ICLR**)*, 2025.

[9] Tommaso Mencattini*, **RC***, and Francesco Locatello. Exploratory Causal Inference in SAEnce. *Proceedings in the Fourteenth International Conference on Learning Representations (**ICLR - top 1%, oral**)*, 2026.

[10] **RC**, Frank Otchere, Nyasha Tirivayi, Gustavo A. Tagliaferro, Falco J. Bargagli-Stoffi, and Francesco Locatello. From Tokens to Policy: Causal and Interpretable Heterogeneous Treatment Effects Identification. In *[**under review**]*, 2026.

Workshop Articles

[11] Yuejiang Liu, **RC**, and Alexandre Alahi. Towards Robust and Adaptive Motion Forecasting: A Causal Representation Perspective. In ***NeurIPS** Workshop on Distribution Shifts: Connecting Methods and Applications*, 2021.

[12] Mauricio Tec, **RC**, Francesca Dominici, and Corwin Zigler. Projecting the climate penalty on $PM_{2.5}$ pollution with spatial deep learning. In ***ICLR** Workshop in Tackling Climate Change with Machine Learning*, 2023.

[13] **RC**, Lukas Lindorfer, Sylvia Cremer, Cordelia Schmid, and Francesco Locatello. Smoke and mirrors in causal downstream tasks. ***ICML**, Workshop in AI for Science: Scaling in AI for Scientific Discovery*, 2024.

[14] Dingling Yao, Dario Rancati, **RC**, Marco Fumero, and Francesco Locatello. Unifying Causal Representation Learning with the Invariance Principle. ***NeurIPS** Workshop on (i) Causal Representation Learning and (ii) UniReps*, 2024.

[15] **RC**, Ilker Demirel, Piersilvio De Bartolomeis, Lukas Lindorfer, Sylvia Cremer, Cordelia Schmid, and Francesco Locatello. Causal lifting of neural representations: Zero-shot generalization for causal inferences. ***ICLR** Workshop on (i) Spurious Correlation and Shortcut Learning and (ii) XAI4Science*, 2025.

[16] **RC*** and Christian Internò*. The narcissus hypothesis: Descending to the rung of illusion. ***NeurIPS** Workshop on Evaluating the Evolving LLM Lifecycle*, 2025.

[17] Dingling Yao, Shimeng Huang, **RC**, Kun Zhang, and Francesco Locatello. The third pillar of causal analysis? a measurement perspective on causal representations. ***ICML** Workshop on Scaling Up Intervention Models*, 2025.

[18] **RC**, Frank Otchere, Nyasha Tirivayi, Gustavo A. Tagliaferro, Falco J. Bargagli-Stoffi, and Francesco Locatello. From tokens to policy: Causal and interpretable heterogeneous treatment effects identification. In ***AISTATS** causality workshop and **ICML** MechInt workshop **(spotlight - top 2%)***, 2026.

* Co-first authors.

Software Packages

[a] Naeem Khoshnevis, **RC**, Daniela Maria Garcia, Kwonsang Lee, Falco Joannes Bargagli Stoffi, "CRE: R Package Causal Rule Ensemble Algorithm", CRAN, 2023 (16 000+ downloads, Website, Github).

[b] **RC**, Naeem Khoshnevis, Falco Joannes Bargagli Stoffi "pycre: Python Package Causal Rule Ensemble Algorithm", PyPI, 2023 (Github).

Events

Selected Talks

Exploratory Causal Inference in SAEnce (oral) **@ICLR'26**
Causality for Science Workshop (invited talk+panel) **@NeurIPS'25**
Exploratory Causal Inference **@CIML'26**
Causality in the era of AI **@Huawei x IHES**
Causality in the era of Foundation Models **@Causality Workshop**
Artificial Causal Inference **@University of Maputo**
Causal Rule Ensemble **@Harvard - School of Public Health**

Summer Schools

M^2L Summer School: Milan 2020-22 (online)
Neurosymbolic Programming: Los Angeles 2022
Causality: Barbados 2025-7, Cambridge 2026 (2x)
MLSS: Senegal 2025

Reviewer

NeurIPS (top reviewer), ICLR, ICML, AAAI, AISTATS
Other Workshops and minor journals

Other Interests

Sport: Marathon runner (2:42:35) @VRC, long-distance hiker, skater and wannabe surfer,
Volunteer: NeurIPS and ICLR (logistic), LeadTheFuture (mentoring), Africatlethics (teaching and coaching), BrixiAmaTe (teaching), CARITAS (childcare), Operazione Mato Grosso (various), AVIS (blood donor).